EE596H: Quiz 1

Total Questions: 1, Total Marks: 20, Time: 60 Mins

1. Consider the following optimization problem

$$
\begin{aligned}
\min \quad & x_1^4 - 2x_2^2 - x_2 \\
\text{s.t.} \quad & x_1^2 + x_2^2 + x_2 \leq 0
\end{aligned}
$$

(a) Solve the KKT conditions for the above problem. [10]

(b) Is the problem convex? [3]

(c) Develop the second order sufficient conditions so as to confirm the minimizer points. [7]

Total: 20

EE596H (Optimization Techniques): Half Semester Examination
Total Questions: 6, Total Marks: 60, Time: 180 Mins

1. Is the following set convex?

$$S = \left\{ \mathbf{x} \in \mathbb{R}^2 : x_1^2 - x_2^2 + x_1 + x_2 \leq 4 \right\}.$$

$\boxed{5}$

2. Verify whether the following function is convex or concave or neither.

$$f(\mathbf{x}) = 4x_1^2 + 3x_2^2 + x_3^2 - 6x_1x_2 + x_1x_3 - \frac{x_1}{2} - 2x_2 + 15.$$

$\boxed{10}$

3. Consider the following problem:

$$\begin{aligned} \text{minimize} \quad & x_1^2 + x_2^2 \\ \text{subject to} \quad & 4 - x_1 - x_2^2 \leq 0 \\ & 3x_2 - x_1 \leq 0 \\ & -3x_2 - x_1 \leq 0 \end{aligned}$$

(a) Solve the above problem graphically. $\boxed{5}$

(b) Verify that KKT conditions are satisfied at the solution points. $\boxed{5}$

(c) Examine the SOSC. [See the snapshot from "Nonlinear Programming" by Bertsekas] $\boxed{5}$

(d) Is the above problem a convex optimization problem? $\boxed{2}$

Total: 17

Proposition 4.3.2: (Second Order Sufficiency Conditions) Consider problem (ICP), assume that f, h, and g are twice continuously differentiable, and let $x^* \in \Re^n$, $\lambda^* \in \Re^m$, and $\mu^* \in \Re^r$ satisfy

$$\nabla_x L(x^*, \lambda^*, \mu^*) = 0, \qquad h(x^*) = 0, \qquad g(x^*) \leq 0,$$

$$\mu_j^* \geq 0, \qquad j = 1, \ldots, r,$$

$$\mu_j^* = 0, \qquad \forall\, j \notin A(x^*),$$

$$y' \nabla_{xx}^2 L(x^*, \lambda^*, \mu^*) y > 0,$$

for all $y \neq 0$ such that

$$\nabla h_i(x^*)' y = 0, \ \forall\, i = 1, \ldots, m, \qquad \nabla g_j(x^*)' y = 0, \ \forall\, j \in A(x^*). \tag{4.54}$$

Assume also that

$$\mu_j^* > 0, \qquad \forall\, j \in A(x^*). \tag{4.55}$$

Then x^* is a strict local minimum of f subject to

$$h(x) = 0, \quad g(x) \leq 0.$$

4. Consider the following problem:

$$\begin{aligned} \text{maximize} \quad & x_2 \\ \text{subject to} \quad & (x_1+1)^2 + (x_2-1)^2 \leq 1 \\ & (x_1-1)^2 + (x_2-1)^2 \leq 1 \end{aligned}$$

(a) Solve the above problem graphically. $\boxed{5}$

(b) Verify the KKT conditions and justify the findings. $\boxed{5}$

Total: 10

5. Write the dual program of the following primal problem (no need to solve the dual, simply write the dual problem):

$$\begin{aligned} \text{minimize} \quad & x_1^2 + x_2^2 - 10x_1 - 10x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 12 \\ & x_1 - x_2 \leq 6 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

It is claimed that the optimal value of the dual problem is -10. Do you agree with the claim? Justify your answer using the duality theorem. $\boxed{8}$

6. Consider the following problem:

$$\begin{aligned} \text{maximize} \quad & x \\ \text{subject to} \quad & x - \frac{1}{2} \leq 0 \\ & x \in X = \{0, 1\} \end{aligned}$$

(a) Find the dual problem and solve it. $\boxed{5}$

(b) Is there any duality gap? Justify using duality theorem. $\boxed{5}$

Total: 10